SDN MININET PROJECT

NETWORK UTILIZATION MONITOR

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Problem Statement :

Measure and display bandwidth utilization across the network.

Project Expectations :

Collect byte counters; • Estimate bandwidth usage; • Display utilization; • Update periodically

GITHUB LINK :

https://github.com/kubervijayakar04-bot/CN--SDN-mini-project

RESULTS / OUTPUTS :

1) POX Controller Start.

```
kuber@kuber-QEMU-Virtual-Machine:~/pox$ cd ~/pox
python3 ./pox.py forwarding.l2_learning monitor
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
INFO:monitor:Network Utilization Monitor Started (every 5 sec)
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.9
WARNING:version:You're running Python 3.13.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
INFO:openflow.of_01:[00-00-00-00-00-01 1] connected
INFO:monitor:Switch Connected: 1
Task <Timer 1 tid:2> caused an exception and was de-scheduled
Traceback (most recent call last):
  File "/home/kuber/pox/pox/lib/recoco/recoco.py", line 317, in cycle
    rv = t.execute()
  File "/home/kuber/pox/pox/lib/recoco/recoco.py", line 111, in execute
    return self.gen.send(v)
           ~~~~~~~~~~~~~^^^
  File "/home/kuber/pox/pox/lib/recoco/recoco.py", line 1078, in run
    rv = self._callback(*self._args,**self._kw)
  File "/home/kuber/pox/ext/monitor.py", line 21, in request_stats
    connection.send(of.ofp_port_stats_request())
    ~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/kuber/pox/pox/openflow/of_01.py", line 874, in send
    assert isinstance(data, of.ofp_header)
           ~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^
AssertionError
INFO:openflow.of_01:[00-00-00-00-00-01 1] closed
```

2) Mininet Topology

```
kuber@kuber-QEMU-Virtual-Machine:~/pox$ sudo mn --topo single,3 --controller=remote,ip=127.0.0.1,port=6634
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
```

3) pingall Connectivity Test

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
```

4) Traffic Generation (h1 ping h2)

```
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.266 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.054 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.105 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.108 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.132 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.301 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.067 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.116 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.111 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.119 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.124 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.115 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.133 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.122 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.131 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.118 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.120 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.153 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.138 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.111 ms
64 bytes from 10.0.0.2: icmp_seq=21 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=22 ttl=64 time=0.146 ms
64 bytes from 10.0.0.2: icmp_seq=23 ttl=64 time=0.137 ms
64 bytes from 10.0.0.2: icmp_seq=24 ttl=64 time=0.111 ms
64 bytes from 10.0.0.2: icmp_seq=25 ttl=64 time=0.117 ms
64 bytes from 10.0.0.2: icmp_seq=26 ttl=64 time=0.112 ms
64 bytes from 10.0.0.2: icmp_seq=27 ttl=64 time=0.108 ms
64 bytes from 10.0.0.2: icmp_seq=28 ttl=64 time=0.105 ms
64 bytes from 10.0.0.2: icmp_seq=29 ttl=64 time=3.33 ms
64 bytes from 10.0.0.2: icmp_seq=30 ttl=64 time=0.123 ms
64 bytes from 10.0.0.2: icmp_seq=31 ttl=64 time=0.129 ms
```

5) Network Utilization Monitor Started Successfully

```
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